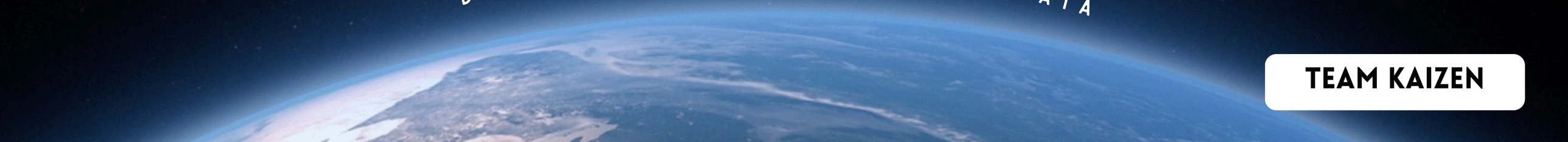
METHANE PULSE LEO

On-Orbit Methane Plume Detection · IBM TerraMind + Sentinel-2

DOWNLINK THE ANSWER. NOT THE DATA



TEAM KAIZEN

WHO PAYS AND WHY RIGHT NOW

The customer : Oil & gas operators, environmental regulators, carbon credit verifiers

Their problem : Methane leaks are often invisible and hard to detect quickly using only ground inspections, especially in landfills, pipelines, and remote industrial sites. Satellites can identify methane related risk signals over large areas, and with onboard AI like TerraMind, they can send instant risk alerts instead of huge raw images, saving time and bandwidth.

The window : Methane disperses fast. Response must be faster than the downlink

80× MORE POTENT

METHANE'S WARMING POWER VS CO_2 OVER 20 YEARS

13 MB → 500 B

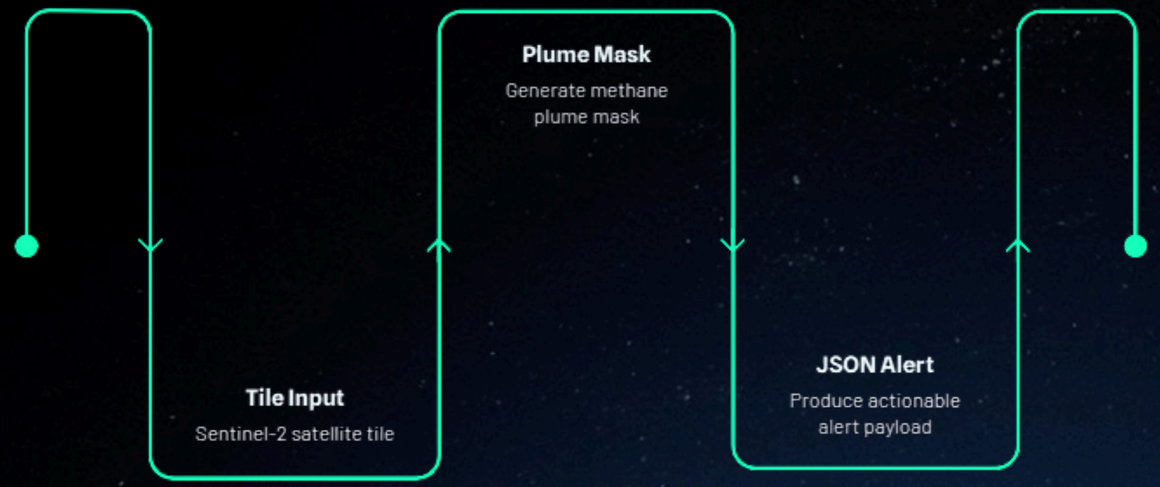
RAW TILE VS. OUR ALERT PACKET

TERRAMIND SEES IT. OUR HEADS ACT ON IT

TERRAMIND SEES IT. OUR HEADS ACT ON IT.	512×512 · SWIR BANDS B11 + B12
TERRAMIND-1.0-SMALL ENCODER	FROZEN · 21.5M PARAMS
FEATURE MAP	14×14 · 384-DIM EMBEDDINGS
DECODERS	SEGMENTATION DECODER → BINARY PLUME MASK SEVERITY CLASSIFIER → LOW / MEDIUM / HIGH
JSON ALERT PACKET	~500 BYTES

WHAT IT ACTUALLY DOES

Demo clip



One tile in. One structured alert out. Under 3 seconds.


```
{
  "STATUS": "ALERT",
  "PLUME_DETECTED": TRUE,
  "COVERAGE_PCT": 0.544,
  "CONFIDENCE": 0.965,
  "SEVERITY": "HIGH",
  "CENTROID": [844340, 2844820]
}
```


Geographic coordinates. Ready for dispatch.

THE EVIDENCE

Model	F1 Score	IoU
Physics baseline (z-score, no ML)	0.208	0.121
TinyUNet (deep learning, no foundation model)	0.658	0.490
MethanePulse-LEO (TerraMind)	~0.52	0.330

2.5× BETTER F1 THAN PHYSICS BASELINE

WHERE WE ARE HONEST & WHERE WE GO NEXT

CURRENT LIMITS

- TerraMind IoU 0.330 < TinyUNet 0.490 , frozen encoder, coarse 14×14 feature map
- Severity accuracy 39.5% proxy labels, not ground-truth emission rates
- Jetson latency not yet benchmarked
- Cloud-free Sentinel-2 only

NEXT 30 DAYS

- Unfreeze last 2 TerraMind blocks → target IoU > 0.50
- Use emission rate from dataset filenames as real severity signal
- Benchmark on Jetson Orin NX (latency + watt draw)
- Extend to SAR for cloud-penetrating detection
- REST API wrapper for OrbitLab integration
- add a classifier to identify the exact methane source, not just general risk